



TEAM ASTROLOGIC Presents

Megacity Climate Intelligence

Offline-proof deterministic climate analytics tracking urban heat islands and nocturnal anomalies app for a better world and fight against climate change



TEAM ASTROLOGIC

Region: Dhaka, Bangladesh

Project : Megacity Climate Intelligence



TEAM MEMBERS & INSTITUTIONS

Tanvir Hosen Nishat (Team Lead) AIUB

Hosni Rabbani BRACU

Mansura Jannat Monima AIUB

Md. Abu Naif Siam BRACU

Md. Farhan Sikdar NSU

Ishtiaque Mahmud Sakif AIUB



Create Empathy For The Problem

Why is it important?

Zero Night Cooling: Severe nocturnal heat halts natural cardiovascular recovery.

Masked Microclimates: Regional assessments overlook localized heat island acceleration.

Humanize it — Who does it affect?

Informal Settlers & Outdoor Laborers

High-Density Settlements: Rickshaw pullers and laborers lack active indoor cooling.

Information Disconnect: Offline workers lack smartphones to access civic warning alerts.

Why is it an opportunity? (Killer data point):

The Opportunity: NASA Empirical Baselines

Decadal Observation: Multi-decade NASA data provides unassailable empirical reality.

Deterministic Action: Transforms theoretical climate debates into life-saving policy.

Data Source We Are Thinking to Using Initially

NASA POWER Daily Point API (T2M_MIN, T2M_MAX, T2M, PRECTOTCORR) for climate anomaly time-series, urban heat island differentials, and regional ecological baseline comparisons.

Solution Approach & Scientific Validity

The Platform:

Megacity Climate Intelligence (MCI) — an open-access, offline-proof climate intelligence engine built on modular city architecture.

Scientific Rigor (Validity):

Evaluates multi-decade NASA observations using tested statistical **functions**:

F1 : Mann-Kendall Test: Establishes true trend direction and statistical significance (p-value).

F2 : Theil-Sen Slope: Calculates the exact rate of urban heat widening per year.

Transparent Evidence:

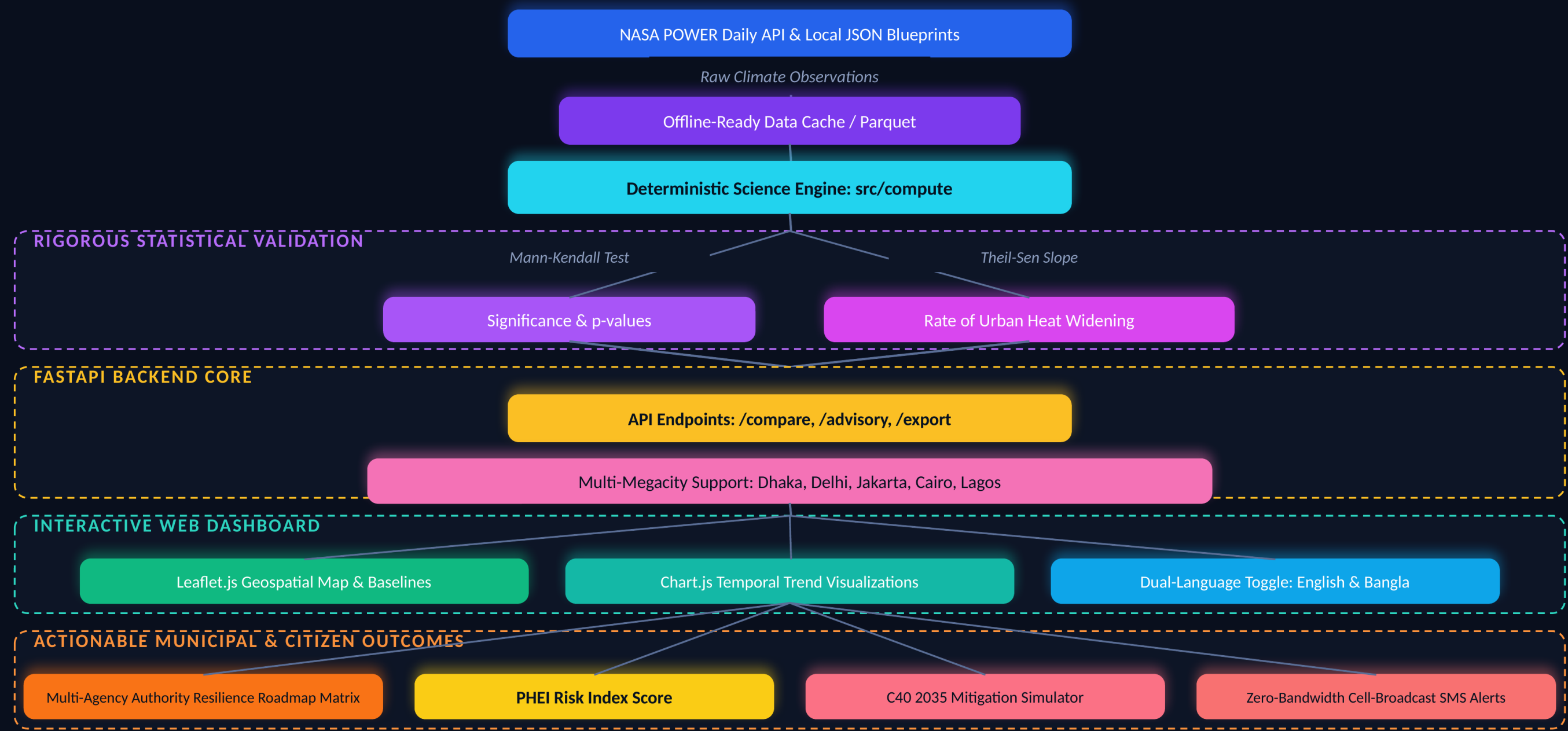
Features an integrated Provenance Drawer showing raw sensor observations, confidence intervals, and processing stages directly to judges.



How Does it Work ?

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System Architecture





NASA SPACE APPS CHALLENGE 2026 • BE AN EARTH SYSTEM TREND DETECTIVE!

Evidence & images



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Megacity Climate Intelligence 🌐 🔍

নাসা স্পেস অ্যাপস ২০২৬ — গ্লোবাল মেগাসিটি ব্লুপ্রিন্ট (টিম অ্যাস্ট্রোলজিক)

🌐 দৃষ্টিকোণ / ব্যবহারকারীর ধরন নির্বাচন:

🌐 সম্পূর্ণ ওভারভিউ

👤 সাধারণ নাগরিক মোড

🏠 নগর পরিকল্পনাবিদ মোড

🔍 বিজ্ঞানী ও বিচারক মোড

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🌐 আন্তর্জাতিক মেগাসিটি নির্বাচন:

Dhaka (Bangladesh) ▼

ক্লাইমেট ভ্যারিয়েবল:

নাইট-টাইম সর্বনিম্ন তাপমাত্রা (T2M_MIN) ▼

ঋতুভিত্তিক বিভাজন:

সম্পূর্ণ ক্যালেন্ডার বছর (জানু - ডিসে) ▼

শুরুর বছর: 2004

ট্রেন্ড বিশ্লেষণ করুন

জনসংখ্যা-ভারযুক্ত তাপ ঝুঁকি সূচক (PHEI)

43 / 100 MODERATE

মোট ঝুঁকিপূর্ণ নগর জনসংখ্যা

23.2 মিলিয়ন

উচ্চ ঝুঁকিপূর্ণ বসতি ও টিনের ঘরের জনগোষ্ঠী

8.82 মিলিয়ন (38%)

জনসংখ্যার ঘনত্ব (প্রতি বর্গ কিমি)

30,093 / km²

🏗️ C40 ক্লাইমেট সিমুলেটর ও নীতি মডেলিং (২০৩৫):

পূর্বাভাস বজায় রাখা (পলিসি পদক্ষেপ ছাড়া) ▼

☒ 🌳 ইন-সিটু আবহাওয়া স্টেশনের সাথে নির্ভুলতা যাচাই

📍 লক্ষ্য মেগাসিটি: Dhaka (Bangladesh) | Lat: 23.8103°, Lon: 90.4125°

🌳 গ্রামীণ নিয়ন্ত্রণ বেসলাইন: Sylhet Rural Baseline (Natural Canopy & High Vegetation Control)

📱 সাধারণ জনগণের জন্য আর্লি ওয়ার্নিং সিস্টেম (স্বয়ংক্রিয় এসএমএস বার্তা)

স্মার্টফোন বা ইন্টারনেটবিহীন শ্রমজীবী ও সাধারণ মানুষের জীবন বাঁচানোর জন্য সরাসরি সেল-ব্রডকাস্ট।

প্রেরক: BTRC / DNCC Climate Alert

কভারেজ আওতা: Grameenphone, Banglalink, Robi, Teletalk



Evidence & images



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Megacity Climate Intelligence

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NASA Space Apps 2026 — Global Megacity Blueprint (Team AstroLogic)

বাংলা

Export Science Bundle

Offline Ready

Select Perspective / Role:

Overview (All)

Citizen Mode

City Planner Mode

Scientist / Judge Mode

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Dhaka (Bangladesh) Urban Core

Lat: 23.8103°, Lon: 90.4125°

Dense Impervious Megacity

✕

Global Blueprint City:

Dhaka (Bangladesh)

Metric:

Night-time Min Temperature (T2M_MIN)

Season Decomposition:

Full Calendar Year (Jan - Dec)

Start Year: 2004

Detect Trends

POPULATION-WEIGHTED HEAT EXPOSURE INDEX (PHEI)

43 / 100 MODERATE

TOTAL URBAN POPULATION EXPOSED

23.2 Million

HIGH-VULNERABILITY SLUM / TIN-ROOF DWELLERS

8.82 Million (38%)

URBAN BUILT-UP DENSITY

30,093 / km²

C40 Urban Heat Mitigation Simulator (2035):

Business-As-Usual (No Policy Action)

☒ Cross-Validate with In-Situ Meteorological Ground Truth

Target Core: Dhaka (Bangladesh) | Lat: 23.8103°, Lon: 90.4125°

Control Baseline: Sylhet Rural Baseline (Natural Canopy & High Vegetation Control)

📱

Low-Bandwidth Citizen Early Warning System (Live SMS Broadcast)

Designed for 23+ million citizens, rickshaw pullers, and informal workers with zero smartphone access.

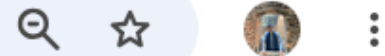
From: BTRC / DNCC Climate Alert

Target Coverage: Grameenphone, Banglalink, Robi, Teletalk



Evidence & images

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Low-Bandwidth Citizen Early Warning System (Live SMS Broadcast)

Designed for 23+ million citizens, rickshaw pullers, and informal workers with zero smartphone access.

YELLOW ADVISORY

From: BTRC / DNCC Climate Alert

MODERATE HEAT: Stay hydrated across Dhaka (Bangladesh). Limit prolonged unshaded exposure during afternoon hours.

Dial: *16100# (Free Municipal Hotline)

Target Coverage: Grameenphone, Banglalink, Robi, Teletalk

Global Standard: UN Early Warnings for All (EW4All) & ITU Protocol

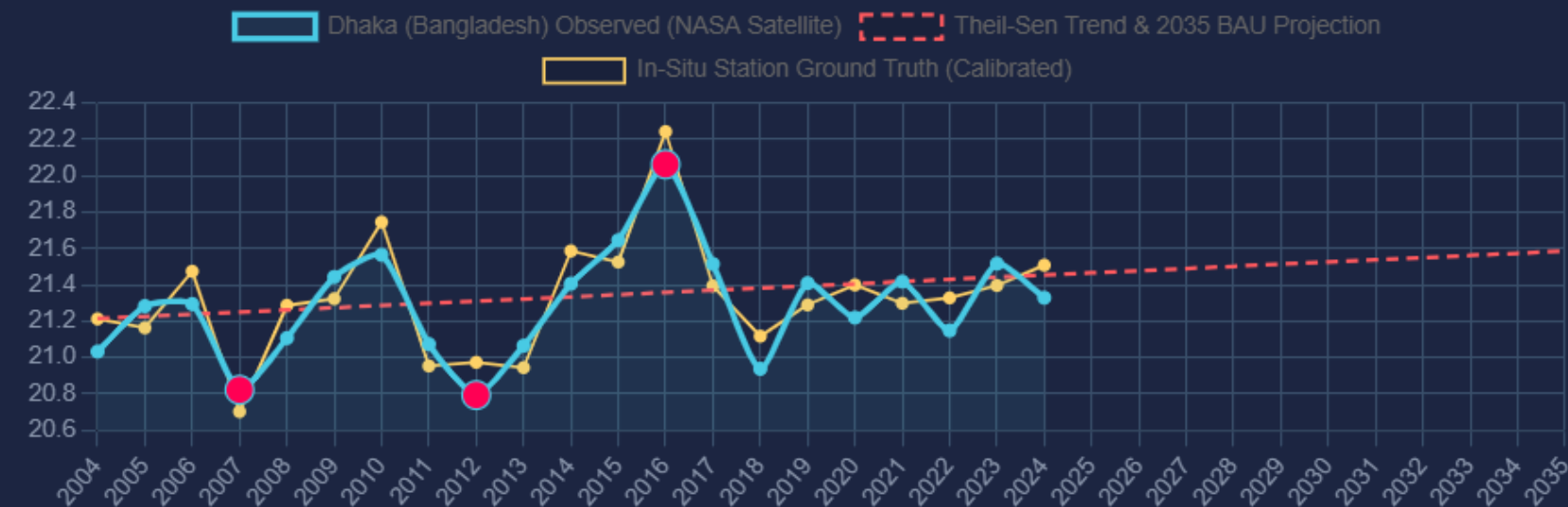
Action Trigger: Automated NASA Night-time Anomaly Threshold (T2M_MIN $\geq 25^{\circ}\text{C}$)

Dhaka (Bangladesh)

Direction: INCREASING

Theil-Sen Slope: 0.0120 /year

P-Value: 0.19413 (Not Significant at $\alpha=0.05$)



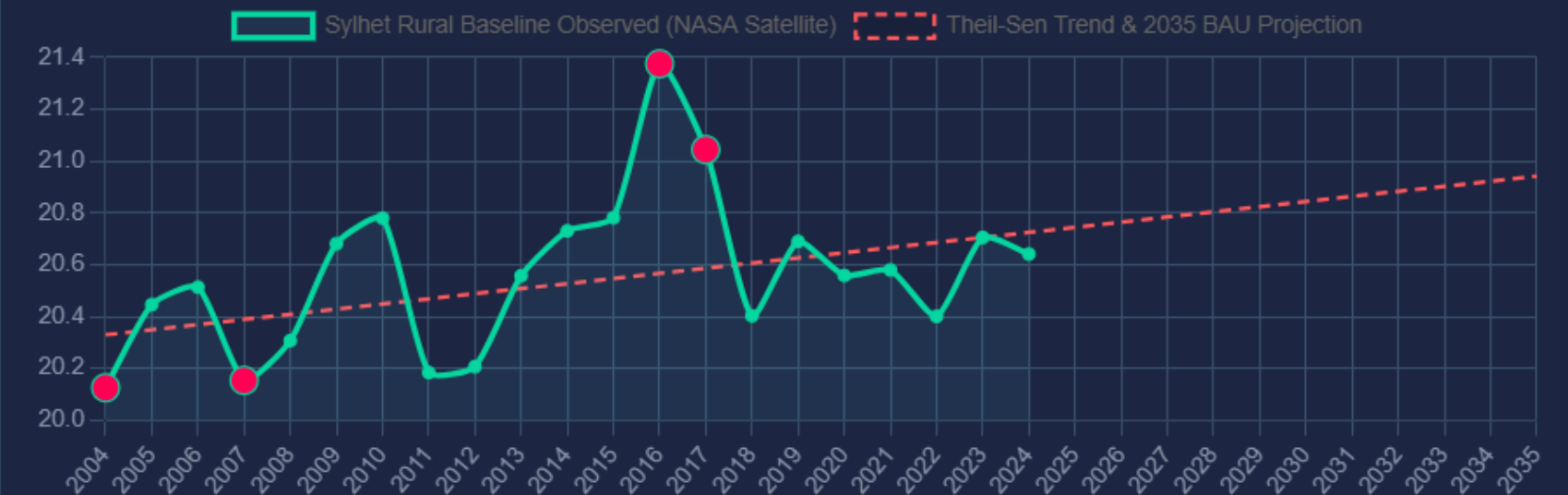
Toggle Provenance Drawer

Sylhet Rural Baseline

Direction: INCREASING

Theil-Sen Slope: 0.0198 /year

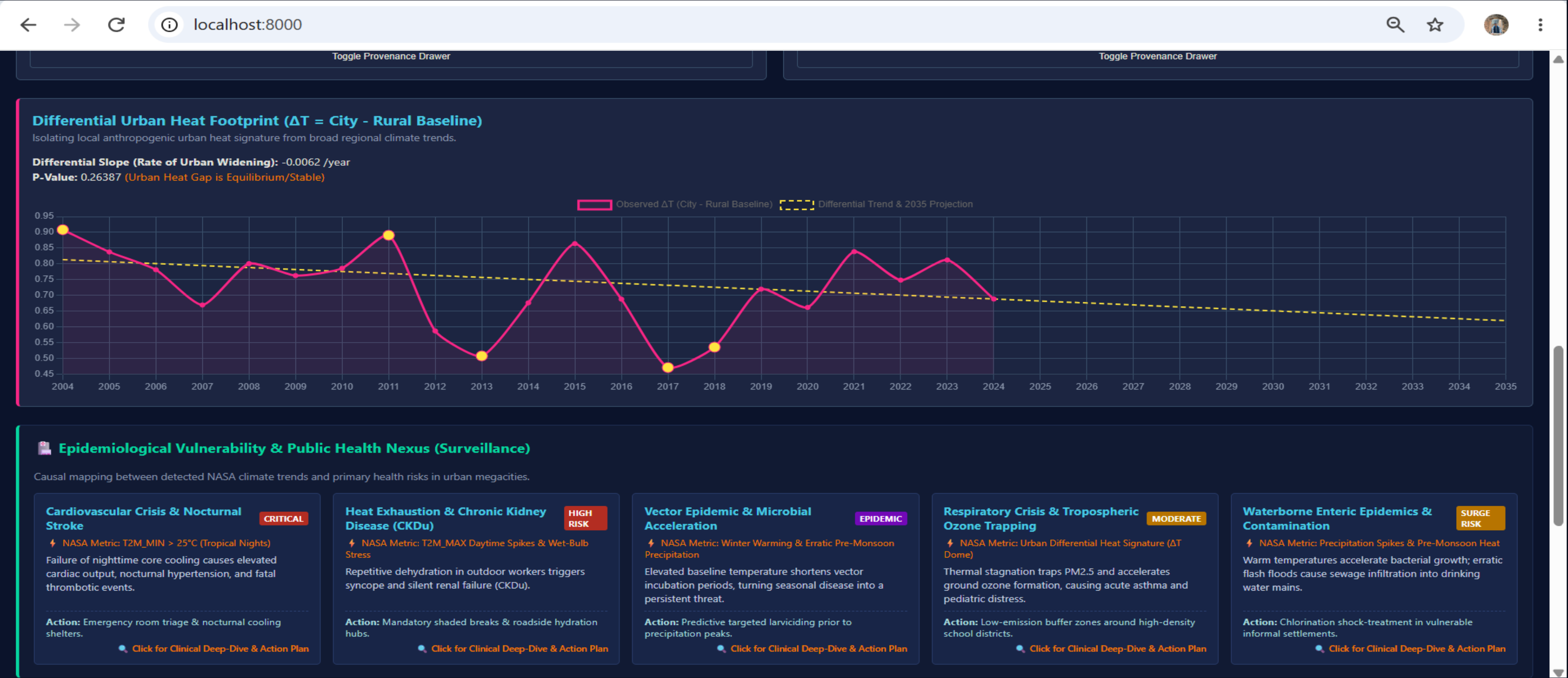
P-Value: 0.04967 (Statistically Significant)



Toggle Provenance Drawer



Evidence & images





Evidence & images

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Action: Emergency room triage & nocturnal cooling shelters.
[Click for Clinical Deep-Dive & Action Plan](#)

Action: Mandatory shaded breaks & roadside hydration hubs.
[Click for Clinical Deep-Dive & Action Plan](#)

Action: Predictive targeted larviciding prior to precipitation peaks.
[Click for Clinical Deep-Dive & Action Plan](#)

Action: Low-emission buffer zones around high-density school districts.
[Click for Clinical Deep-Dive & Action Plan](#)

Action: Chlorination shock-treatment in vulnerable informal settlements.
[Click for Clinical Deep-Dive & Action Plan](#)

Scientific Interpretation & Trend Attribution

Evaluation Context: Megacity: Dhaka (Bangladesh) | Metric: T2M_MIN | Season: ALL

- **Regional Rural Baseline (Sylhet Rural Baseline):** Shows a slope of **0.0198/year** with $p = 0.04967$. Reflects natural regional background warming.
- **Urban Megacity Core (Dhaka (Bangladesh)):** Shows a slope of **0.0120/year** with $p = 0.19413$.
- **Urban Residual Footprint (ΔT):** By subtracting rural baseline, the differential slope is **-0.0062/year** ($p = 0.26387$). The differential remains balanced with macro regional movements.

Authority Resilience Roadmap: Data-to-Decision Matrix
Phased multi-agency interventions triggered by detected climate trends.

Short-Term (0-1 Yr)

Mid-Term (3-5 Yrs)

Long-Term (10-15 Yrs)

Emergency Heat Protocols (DGHS & icddr,b)

- **Automated Pre-Alerts:** Pre-stage cold IV fluid reserves in regional hospitals 48 hours prior to forecasted anomalies.
- **Nocturnal Public Shelters:** Open air-conditioned public community centers at night for high-density informal workers.
- **Midday Labor Curtailment:** Mandate hourly 15-minute shaded cooling pauses between 11 AM - 3 PM during active heat spikes.

Immediate Hydration & Mist Infrastructure (Dhaka WASA)

- **Intersectional Hydration:** Deploy automated cool potable water refill stations and free ORS at primary transit hubs.
- **Mobile Water Misting:** High-volume water vaporization along dense corridors to suppress ambient temperatures.
- **Shock-Chlorination:** Immediate chlorine dosing in municipal supply lines following erratic precipitation spikes.

Targeted Vector Precision (DNCC & DSCC)

- **Data-Triggered Larviciding:** Automated dispatch of biological larvicides exactly 72 hours after NASA rainfall anomalies.
- **Rooftop Drone Audits:** Thermal drone surveillance over stagnant pools in construction sites and rooftop gardens.

Discuss Applications

1 Spatial Detection

Contrasts megacities against regional baselines (e.g., Dhaka vs. Sylhet) to isolate anthropogenic heat domes.

Dhaka



Delhi



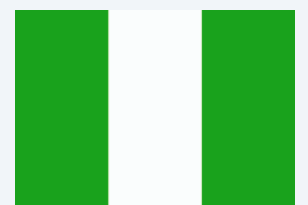
Jakarta



Cairo



Lagos



2 PHEI Risk Score

Calculates human vulnerability indices by weighting density against informal slum structures.

3 Zero-Bandwidth Alerts

Simulates how cool roofs and urban canopy programs bend 2035 municipal warming curves.

4 c40 2035 Simulator

Calculates human vulnerability indices by weighting density against informal slum structures.

5 Global Blueprint

Modular configuration enables instant deployment across Dhaka, Delhi, Jakarta, Cairo, and Lagos.



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Live Demo Video (For Visualization)

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Megacity Climate Intelligence 🌍 🔍

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English 🇬🇧 অফলাইন প্রস্তুত

👁️ দৃষ্টিকোণ / ব্যবহারকারীর ধরন নির্বাচন:

🌐 সম্পূর্ণ ওভারভিউ

👤 সাধারণ নাগরিক মোড

🏙️ নগর পরিকল্পনাবিদ মোড

🔬 বিজ্ঞানী ও বিচারক মোড

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🌐 আন্তর্জাতিক মেগাসিটি নির্বাচন: Dhaka (Bangladesh) ▼

📅 ক্লাইমেট ভ্যারিয়েবল: নাইট-টাইম সর্বনিম্ন তাপমাত্রা (T2M_MIN) ▼

🗺️ খাত্ত্বভিত্তিক বিভাজন: সম্পূর্ণ ক্যালেন্ডার বছর (জানু - ডিসে) ▼

📅 শুরুর বছর: 2004 🔴

📊 ট্রেন্ড বিশ্লেষণ করুন

জনসংখ্যা-ভারযুক্ত তাপ ঝুঁকি সূচক (PHEI)
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মোট ঝুঁকিপূর্ণ নগর জনসংখ্যা
23.2 মিলিয়ন

উচ্চ ঝুঁকিপূর্ণ বস্তি ও টিনের ঘরের জনগোষ্ঠী
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🔧 C40 ক্লাইমেট সিমুলেটর ও নীতি মডেলিং (২০৩৫): পূর্বাবস্থা বজায় রাখা (পলিসি পদক্ষেপ ছাড়া) ▼

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🌿 গ্রামীণ নিয়ন্ত্রণ বেসলাইন: Sylhet Rural Baseline (Natural Canopy & High Vegetation Control)



Impact and Your Needs

What will this idea change?

Proactive Thermal Governance

Deterministic Planning: Replaces reactive emergency relief with predictive microclimate mitigation.

Targeted Cooling: Empowers cities to pinpoint nocturnal hotspots before fatal heat waves peak.

Universal Inclusive Alerting

Zero-Data Broadcast: Triggers automated cell-broadcast SMS without smartphone requirements.

Toll-Free Helpline: Directly links outdoor laborers to emergency guidance via *16100#.

C40 2035 Mitigation Simulator

Pre-Capital Validation: Simulates temperature drops for cool roofs and canopy expansion before budget spend.

ROI Confidence: Equips municipal leaders with verified evidence for green infrastructure financing.



Impact and Your Needs

What is our 'burning platform'?

Municipal City Partnerships

Civic Integration: Direct mandates with Dhaka North/South City Corporations and Cairo Governorate.

Official Protocols: Embedding the Data-to-Decision matrix into statutory municipal emergency workflows.

Telecommunication Gateways

Carrier Agreements: Partnerships with national mobile operators for zero-rated emergency cell broadcasts.

Automated Triggering: Real-time delivery whenever NASA models identify high-risk thermal anomalies.

Ground Micro-Weather Mesh

Hyperlocal Ground Truth: Integrating IoT street sensors to continuously cross-validate satellite data.

Feedback Calibration: Refining remote thermal retrievals against localized street-level canopies and alleys.

Thank You For Your Attention !

Megacity Climate Intelligence (MCI)
Team : AstroLogic

